

Overall Merit: Strong accept

Summary The paper asks how far semiconductor integrated circuits can scale in component count per chip (and thus cost/performance), and what technical/economic constraints will matter as integration grows.

Moore's central mechanism is an economic optimum: cost per component falls roughly inversely with component count for small chips, but eventually yield loss dominates and pushes cost up, which produces a "minimum cost per component" at a particular complexity level. He argues that this optimum shifts upward rapidly, implying exponential growth in economically optimal integration.

The main scaling claim: minimum-cost complexity increases by roughly a factor of two per year, which at that time, projecting to around 65,000 components per chip by 1975. Surprisingly, this prediction held up well for the next 40 years.

Strengths

1. The author made concrete, falsifiable prediction. Even if approximate, the "factor of two per year" and "65,000 components by 1975" claims are explicit enough to motivate roadmapping and debate.
2. The minimum-cost argument (inverse cost per component vs yield-driven rise) gives a concrete rationale for why industry would pursue higher integration, not just why it's physically possible.
3. There are some attention to physical constraints, not just optimism. Heat and analog limitations are at least addressed with specific reasoning.

Weaknesses

1. The paper leans heavily on qualitative extrapolation and mentions graphs without fully specifying model form, data sources, or sensitivity.
2. There are overconfident assumptions around manufacturability/yield. The claim that there is "no fundamental obstacle" to 100% yields and "only engineering effort is needed" is quite bold.

Detailed Comments I liked that the paper separates "can we build it?" from "should we build it?" and explicitly frames the latter as an optimization over engineering amortization and automation. It is surprising that the scaling claims held up so well for so long. As a reviewer, I'd ask for a clearer explanation of the underlying data behind the "factor of two per year" trend.

Discussion Questions

1. Is the most important contribution the exponential trend claim (doubling) or the economic mechanism (minimum cost per component)?
2. The paper argues heat is manageable because ICs are effectively 2D and shrinking reduces capacitance. Is this a valid assumption?